Lecture 40 (Hinglish): Deep Learning (Part-5)

CIFAR-10 CNN, NLP Foundations, Word Embeddings, RNN & LSTM

Yeh **Lecture 40: Deep Learning (Part-5)** ka complete comprehensive master guide hai. Is lecture me Computer Vision ke practical PyTorch project se lekar Natural Language Processing (NLP), Word Embeddings, Recurrent Neural Networks (RNN), aur Long Short-Term Memory (LSTM) networks ko detail me cover kiya gaya hai.

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1. Directory Structure

```
lecture_40/
├── Lecture_Code/
│   ├── CNN_for_CIFAR10.ipynb             <- Complete executable PyTorch notebook
│   └── Resources/
│       ├── 5 Popular CNN Architectures.pdf <- Landmark CNN reference
│       ├── Word2Vec Reading Material.png  <- Vector semantics reference
│       └── LSTM Reference Pages           <- Hochreiter & Schmidhuber reference
│
├── Notes_40.01/  <- CIFAR-10 Dataset & PyTorch Preprocessing Pipeline
├── Notes_40.02/  <- Building the 3-Block Deep CNN Architecture in PyTorch
├── Notes_40.03/  <- CNN Training, Optimization & Test Evaluation on CIFAR-10
├── Notes_40.04/  <- Landmark CNN Architectures: From LeNet-5 to ResNet
├── Notes_40.05/  <- Foundations of Natural Language Processing (NLP) & Text Prepr
├── Notes_40.06/  <- Text Representation: One-Hot Encoding, Bag-of-Words (BoW) & N
├── Notes_40.07/  <- Advanced Text Representation: TF-IDF & Distributed Word Embed
├── Notes_40.08/  <- Recurrent Neural Networks (RNN): The Recurrent Loop & Time Un
├── Notes_40.09/  <- Backpropagation Through Time (BPTT), Vanishing & Exploding Gr
├── Notes_40.10/  <- Long Short-Term Memory (LSTM) Networks: Cell State & Gating A
│
└── Practice_Problems/
    ├── README.md                         <- 15 High-Yield Technical Problems (En
    ├── README_HINGLISH.md                <- 15 High-Yield Technical Problems (Hi
    ├── Solutions.md                      <- Full Mathematical & Code Solutions (
    └── Solutions_HINGLISH.md             <- Full Mathematical & Code Solutions (
```

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2. Submodules Overview

1. **Notes_40.01:** CIFAR-10 dataset ($60,000$ images, 10 classes), PyTorch `DataLoader` ($B = 64$, 782 batches), Normalization pipeline $[-1, 1]$.

2. **Notes_40.02:** PyTorch me 3-Block Deep CNN build karna. Spatial reduction ($32 \rightarrow 16 \rightarrow 8 \rightarrow 4$), 2048 flattened features, aur $620,362$ parameters calculation.
3. **Notes_40.03:** Training loop, `CrossEntropyLoss`, Adam optimizer, `model.eval()`, `torch.no_grad()`, aur $72\% - 75\%$ test accuracy evaluation.
4. **Notes_40.04:** Landmark CNNs: LeNet-5 se ResNet tak. VGG factorization (3×3 stacking) aur ResNet skip connections $\mathbf{y} = \mathcal{F}(\mathbf{x}) + \mathbf{x}$.
5. **Notes_40.05:** NLP Foundations: Syntax vs Semantics, Tokenization, Stopwords removal, Stemming vs Lemmatization.
6. **Notes_40.06:** Text Representation: One-Hot encoding ki sparsity, Bag-of-Words Document-Term Matrix (DTM), aur N-Grams negation handling.
7. **Notes_40.07:** TF-IDF statistical weighting formula, Word2Vec dense continuous embeddings, CBOW vs Skip-Gram, aur vector arithmetic.
8. **Notes_40.08:** RNN architecture, recurrent memory loop, time unrolling, parameter sharing, aur Karpathy ke 5 sequence types.
9. **Notes_40.09:** Backpropagation Through Time (BPTT), Jacobian decay, Vanishing Gradients ($|\lambda| < 1$), Exploding Gradients ($|\lambda| > 1$), aur Gradient Clipping.
10. **Notes_40.10:** LSTM Cell Architecture, Constant Error Carousel (CEC), 3 Gates (Forget, Input, Output), Linear Cell State Highway update math, aur PyTorch code.

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3. Notebook-Friendly Hand-Drawn Diagrams

Saare concepts ke liye hand-drawn notebook-style diagrams (`Comic Sans MS`, blue/black pen aesthetic, clean mathematical annotations) use kiye gaye hain. Instructor face/webcam images ko completely remove kar diya gaya hai.